

ORIGINAL ARTICLE

Infrastructure determinants of urban-rural disparities in specialized cardiac care in Kazakhstan: A cross-sectional analytical study

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ABSTRACT

Background and aim: Urban-rural disparities in cardiovascular healthcare accessibility remain an important public health problem in Kazakhstan despite ongoing healthcare reforms and expansion of specialized cardiac services. This study aimed to evaluate infrastructural determinants associated with accessibility and perceived quality of specialized cardiac care among urban and rural populations in the Almaty region of Kazakhstan.

Methods: A cross-sectional analytical study was conducted between September and December 2025 and included 350 patients with cardiovascular diseases and 152 healthcare professionals involved in cardiac care delivery. Data were collected using structured questionnaires assessing accessibility of specialized cardiac services, pharmaceutical supply, rehabilitation availability, digitalization, and perceived quality of care. Urban-rural differences were analyzed using chi-square and Mann-Whitney tests. Logistic and multivariable linear



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regression analyses were performed to identify determinants associated with patient dissatisfaction and professional-perceived quality of care.

Results: Rural patients reported significantly lower accessibility of high-technology medical care compared with urban residents (28.7% vs. 46.7%; $p=0.019$) and lower adequacy of pharmaceutical supply (31.7% vs. 51.1%; $p=0.032$). Insufficient drug supply was the strongest predictor of dissatisfaction with cardiac care ($OR=2.31$; 95% CI: 1.48–3.61; $p<0.001$). Inaccessibility of high-technology medical care ($OR=1.89$; 95% CI: 1.21–2.95; $p=0.005$) and rural residence ($OR=1.62$; 95% CI: 1.08–2.44; $p=0.021$) were also independently associated with dissatisfaction. Healthcare professionals identified digitalization level ($\beta=0.40$; $p=0.017$) and adequacy of material resources ($\beta=0.35$; $p=0.002$) as major determinants of perceived quality of care.

Conclusions: Substantial urban–rural disparities in specialized cardiac care persist in Kazakhstan and are primarily associated with infrastructural and organizational barriers. Improvement of pharmaceutical accessibility, digital integration, rehabilitation services, and continuity of care may represent key priorities for reducing regional inequalities in cardiovascular healthcare delivery. (www.actabiomedica.it)

Key words: cardiovascular diseases, healthcare accessibility, rural health, digitalization, cardiac rehabilitation, healthcare inequality, Kazakhstan

Introduction

Cardiovascular diseases (CVDs) remain the leading cause of mortality worldwide, accounting for approximately 18.6 million deaths annually and nearly 32% of all global deaths (1). According to the Global Burden of Disease 2019 study, ischemic heart disease and stroke remain the two major contributors to cardiovascular mortality and disability worldwide (1). More than 75% of premature cardiovascular deaths occur in low- and middle-income countries, where limited healthcare resources, delayed diagnosis, and unequal access to specialized medical services substantially worsen outcomes (2). In addition to demographic ageing, the increasing prevalence of obesity, diabetes mellitus, hypertension, smoking, and sedentary lifestyle continues to accelerate the global cardiovascular burden (1,2).

Central Asia remains one of the regions with persistently high cardiovascular morbidity and mortality (3). In Kazakhstan, cardiovascular diseases continue to occupy the leading position in the national mortality structure despite ongoing healthcare reforms and modernization programs (4,5). National epidemiological

analyses demonstrated that the standardized mortality rate from cardiovascular diseases decreased from 178.9 per 100,000 population in 2016 to 163.1 per 100,000 in 2019; however, during the COVID-19 pandemic mortality increased sharply and reached 226.9 per 100,000 population in 2021 (4).

Urban–rural disparities in cardiovascular healthcare accessibility represent a major public health problem worldwide (6–8). International studies demonstrate that rural populations experience delayed emergency medical response, lower accessibility of high-technology cardiac interventions, reduced participation in cardiac rehabilitation programs, and higher long-term mortality following acute cardiovascular events (7–10). Structural barriers contributing to these disparities include geographical remoteness, transportation limitations, shortages of cardiologists and specialized personnel, insufficient diagnostic capacity, instability of pharmaceutical supply, and fragmentation of referral pathways (6–9). A systematic review by Alanazy et al. demonstrated that delayed emergency medical service accessibility in rural regions is associated with poorer cardiovascular outcomes and increased mortality risk (9). In addition, Taylor et al.

reported that participation in cardiac rehabilitation programs significantly improves long-term cardiovascular outcomes, although access to such services remains substantially lower in rural populations (10).

These challenges are particularly relevant for Kazakhstan because of the country's large geographical territory and heterogeneous distribution of healthcare infrastructure (12). More than 40% of the national population resides in rural areas, where accessibility of specialized cardiac services remains limited (12). Previous studies conducted in Kazakhstan demonstrated persistent urban-rural differences in cardiovascular morbidity, availability of specialized medical care, and healthcare accessibility (5,6). Rural healthcare organizations frequently experience shortages of cardiologists, limited availability of advanced diagnostic equipment, restricted access to high-technology medical care, and delayed referral for specialized interventions (6). Furthermore, continuity of cardiovascular care and rehabilitation services in rural regions remains insufficiently developed (10).

In recent years, Kazakhstan has implemented healthcare modernization programs focused on digitalization, telemedicine, strengthening primary healthcare, and expansion of specialized medical services (11,12). Digital health technologies, including telecardiology and remote monitoring systems, have demonstrated potential for improving continuity of cardiovascular care and reducing geographical barriers to specialized services (11). However, despite ongoing reforms, substantial infrastructural and organizational inequalities between urban and rural healthcare systems persist. According to recent national analyses, cardiovascular morbidity and mortality increased during the pandemic period among both urban and rural populations of Kazakhstan, particularly in northern regions of the country (13). Long-term epidemiological trends additionally demonstrate a sustained increase in chronic cardiovascular disease prevalence associated with population ageing and accumulation of modifiable risk factors (14,15).

Most studies conducted in Kazakhstan have primarily focused on epidemiological trends, prevention strategies, or clinical outcomes in tertiary cardiac centers (4,5). Comprehensive investigations integrating patient-reported accessibility, healthcare professional assessments, infrastructural capacity, pharmaceutical

accessibility, and digitalization indicators within a unified analytical framework remain limited. In particular, insufficient evidence exists regarding whether perceived quality of specialized cardiac care is predominantly associated with clinical expertise or with infrastructural and organizational determinants.

Given the strategic importance of reducing cardiovascular mortality and improving healthcare equity, identification of infrastructure-related determinants affecting accessibility and quality of specialized cardiac care represents an important public health priority for Kazakhstan. Understanding the mechanisms underlying urban-rural disparities may support development of targeted organizational and healthcare policy interventions aimed at improving continuity, accessibility, and quality of cardiovascular services.

Therefore, this study aimed to evaluate urban-rural disparities in specialized cardiac care and to identify infrastructural determinants associated with accessibility and perceived quality of cardiac care in Kazakhstan using the Almaty region as a case study.

Patients and Methods

Study design and setting

A cross-sectional analytical study was conducted between September and December 2025 in the Almaty region of the Republic of Kazakhstan. The study aimed to evaluate accessibility, quality, and infrastructural determinants of specialized cardiac care among urban and rural populations. The Almaty region was selected because it represents a heterogeneous healthcare environment combining highly specialized tertiary cardiology centers with geographically remote rural healthcare facilities characterized by variable organizational and infrastructural capacity. The study was conducted and reported in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) recommendations for cross-sectional studies (16).

Study population

The study included two independent participant groups consisting of patients with cardiovascular diseases and healthcare professionals involved in cardiac

care delivery. The patient population included adults diagnosed with cardiovascular diseases according to the International Classification of Diseases, 10th Revision (ICD-10 codes I00-I99), who had received outpatient or inpatient cardiac care during the study period. Healthcare professionals included cardiologists, general practitioners, nurses, and healthcare administrators directly involved in cardiovascular care provision within participating healthcare institutions.

Eligible patients were aged 18 years or older, had a confirmed cardiovascular diagnosis, and were capable of providing informed consent. Patients with severe cognitive impairment, acute psychiatric disorders, or inability to complete the questionnaire independently were excluded from participation. Healthcare professionals were eligible if they were actively employed in participating healthcare institutions and directly involved in the organization or delivery of cardiac care services during the study period.

Participants were recruited consecutively during routine outpatient consultations and inpatient treatment within participating healthcare organizations. Consecutive sampling was used to minimize selective recruitment and improve representativeness of the study population.

Sample size

The minimum required sample size for the patient population was calculated using a conservative prevalence estimate of 50%, confidence level of 95%, and margin of error of 5%, resulting in an estimated minimum sample size of 323 participants. The final patient sample exceeded the calculated requirement and was considered sufficient for subgroup comparisons and multivariable regression analysis. The healthcare professional sample reflected the staffing structure of participating healthcare institutions and was considered adequate for multivariable modeling and effect size estimation.

Data collection instruments

Data were collected using two structured author-developed questionnaires specifically designed to assess accessibility and quality of specialized cardiac care

in urban and rural healthcare settings. Questionnaire development was informed by the World Health Organization conceptual frameworks on healthcare accessibility and health system performance as well as previously published studies evaluating healthcare access, organizational barriers, and cardiovascular healthcare disparities (17,18).

The patient questionnaire included sections assessing accessibility of cardiac care services, waiting time for specialist consultation, emergency medical service responsiveness, availability of high-technology medical care, sufficiency of pharmaceutical supply, accessibility of rehabilitation services, and overall satisfaction with cardiac care. Responses were recorded using dichotomous, categorical, and five-point Likert-scale formats ranging from “very poor” to “very good”.

The healthcare professional questionnaire evaluated organizational and infrastructural characteristics of cardiac care delivery, including adequacy of material and technical resources, staffing sufficiency, availability of diagnostic equipment, digitalization level, coordination between levels of care, and perceived quality of specialized cardiac services. An integral perceived quality index was calculated using aggregated standardized domain scores, with higher values indicating better perceived quality of care.

Because validated instruments specifically assessing infrastructural determinants of specialized cardiac care accessibility in Central Asian healthcare systems were unavailable, both questionnaires were developed de novo. Content validity was assessed by a multidisciplinary expert panel consisting of cardiologists, public health specialists, epidemiologists, and healthcare management researchers. Face validity and comprehensibility were evaluated during pilot testing involving 20 patients and 15 healthcare professionals who were not included in the final analysis. Minor linguistic and structural modifications were introduced following pilot assessment.

The questionnaires were developed simultaneously in Kazakh and Russian languages using forward translation and consensus harmonization procedures to ensure semantic equivalence between versions. Internal consistency of multi-item domains included in the integral quality index demonstrated acceptable

reliability, with Cronbach's alpha coefficients exceeding 0.70.

Data collection procedure

Data collection was performed between January and December 2023 in outpatient clinics, district hospitals, and specialized cardiology centers. Patient questionnaires were administered during routine outpatient or inpatient visits through structured face-to-face interviews conducted by trained researchers. Participants capable of independent completion were allowed to complete questionnaires under investigator supervision. The average completion time for the patient questionnaire ranged from 12 to 15 minutes.

Healthcare professional questionnaires were self-administered in paper format during working hours without participation of supervisors or department heads in order to minimize social desirability bias and administrative influence. The average completion time ranged from 10 to 13 minutes. Prior to data collection, all interviewers underwent standardized training regarding questionnaire administration, informed consent procedures, and confidentiality requirements. Completed questionnaires were checked for completeness immediately after collection in order to minimize missing data and recording errors.

Outcome measures

The primary study outcomes included patient dissatisfaction with cardiac care and the professional-perceived quality index. Patient dissatisfaction was assessed using aggregated responses related to accessibility barriers, adequacy of medical services, availability of resources, and overall satisfaction with cardiac care. The professional-perceived quality index represented a continuous composite measure reflecting healthcare professionals' assessment of organizational and infrastructural quality of cardiac services.

The main explanatory variables included insufficiency of pharmaceutical supply, inaccessibility of high-technology medical care, emergency medical service delay, adequacy of material and technical resources, digitalization level, and rural versus urban residence.

Statistical analysis

Data were entered into Microsoft Excel and analyzed using IBM SPSS Statistics version 26.0 (IBM Corp., Armonk, NY, USA). Double data entry and cross-validation procedures were performed to minimize entry errors and improve data accuracy.

Continuous variables were presented as mean $\pm$ standard deviation for normally distributed data and median with interquartile range for non-normally distributed variables. Categorical variables were summarized using frequencies and percentages. Distribution normality was assessed using the Shapiro-Wilk test together with visual inspection of histograms and Q-Q plots.

Comparisons between urban and rural groups were performed using Pearson's chi-square test for categorical variables, independent samples t-test for normally distributed continuous variables, and Mann-Whitney U test for non-normally distributed variables. Effect sizes were estimated using Cramer's V and Cohen's d where appropriate. Statistical significance was established at $p < 0.05$.

Multiple linear regression analysis was performed to identify determinants associated with the professional-perceived quality index. Independent variables included adequacy of material and technical resources, digitalization level, and organizational coordination indicators. Binary logistic regression analysis was used to identify predictors of patient dissatisfaction with cardiac care. Independent variables included insufficiency of drug supply, inaccessibility of high-technology medical care, emergency medical service delay, and rural residence. Results were reported as odds ratios with 95% confidence intervals.

Regression assumptions including linearity, homoscedasticity, independence of residuals, and multicollinearity were assessed prior to interpretation of regression models. Multicollinearity was evaluated using the variance inflation factor (VIF), with values below 5 considered acceptable. Calibration of the logistic regression model was assessed using the Hosmer-Lemeshow goodness-of-fit test.

Receiver operating characteristic (ROC) analysis was additionally performed to evaluate predictive performance of the logistic regression model using area

under the curve (AUC), sensitivity, and specificity estimates.

Data completeness exceeded 95% across variables. Variables with less than 5% missing observations were analyzed using complete-case analysis. Sensitivity analysis comparing complete-case and available-case datasets demonstrated no substantial differences in effect estimates.

Ethical considerations

The study protocol was approved by the Local Ethics Committee of Kazakhstan Medical University (Protocol #IRB-193-2024 No. 8 dated 06 May 2024). The study was conducted in accordance with the principles of the Declaration of Helsinki. All participants provided written informed consent prior to enrollment. Participation was voluntary, all collected data were anonymized before analysis, and no financial or non-financial incentives were provided.

Results

Characteristics of the study population

A total of 350 patients with cardiovascular diseases and 152 healthcare professionals participated in the study. Among patients, 242 (69.2%) resided in rural areas, while 108 (30.8%) were urban residents. The

mean age of the patient population was 56.4 ± 11.2 years (95% CI: 55.2–57.6), and women accounted for 58.0% (n=203) of the cohort. Healthcare professionals had a mean age of 36.5 ± 6.8 years and included cardiologists, general practitioners, nurses, and healthcare administrators involved in specialized cardiac care delivery (Table 1).

The predominance of rural residents reflected the regional demographic structure and allowed detailed evaluation of accessibility-related disparities between rural and urban populations.

Urban-Rural differences in accessibility of specialized cardiac care

Significant disparities between rural and urban populations were identified across several indicators of healthcare accessibility (Table 2).

Adequate pharmaceutical supply was reported significantly less frequently among rural patients than among urban residents. Full availability of prescribed medications was reported by only 31.7% of rural participants compared with 51.1% of urban participants ($\chi^2=6.87$; $p=0.032$; Cramer's $V=0.18$).

Accessibility of high-technology medical care also differed substantially between groups. Access to advanced cardiac interventions and specialized diagnostic procedures was reported by 28.7% of rural patients compared with 46.7% of urban patients ($\chi^2=7.94$; $p=0.019$; Cramer's $V=0.21$).

Table 1. Sociodemographic characteristics of patients (n=350).

Characteristic	Category	N	% (95% CI)
Sex	Female	203	58.0 (52.8–63.1)
	Male	147	42.0 (36.9–47.2)
Residence	Rural	242	69.2 (64.1–73.9)
	Urban	108	30.8 (26.1–35.9)
Mean age (years)	-	56.4 ± 11.2	95% CI: 55.2–57.6

Table 2. Urban-rural differences in accessibility indicators.

Indicator	Rural %	Urban %	χ^2	p-value	Effect size
Adequate drug supply	31.7	51.1	6.87	0.032	$V=0.18$
Accessibility of HTMC	28.7	46.7	7.94	0.019	$V=0.21$
Timely emergency response	57.4	75.6	5.11	0.078	$V=0.15$

Table 3. Binary logistic regression model for patient dissatisfaction with cardiac care (n=350).

Predictor	OR	95% CI	p-value
Insufficient drug supply	2.31	1.48-3.61	<0.001
Inaccessibility of HTMC	1.89	1.21-2.95	0.005
Rural residence	1.62	1.08-2.44	0.021
Emergency service delay	1.41	0.91-2.18	0.118

Timely emergency medical service response demonstrated a similar trend. Timely emergency care was reported by 57.4% of rural residents and 75.6% of urban residents; however, the difference did not reach conventional statistical significance ($\chi^2=5.11$; $p=0.078$).

The largest disparity was observed for accessibility of high-technology medical care, suggesting that infrastructural and organizational limitations remain concentrated in geographically remote areas.

Predictors of patient dissatisfaction with cardiac care

Binary logistic regression analysis was performed to identify independent predictors associated with dissatisfaction with specialized cardiac care (Table 3).

Insufficiency of pharmaceutical supply demonstrated the strongest independent association with dissatisfaction. Patients reporting inadequate medication availability had more than twofold higher odds of dissatisfaction with cardiac care services (OR=2.31; 95% CI: 1.48-3.61; $p<0.001$).

Inaccessibility of high-technology medical care was also significantly associated with dissatisfaction (OR=1.89; 95% CI: 1.21-2.95; $p=0.005$). Rural residence independently increased the probability of dissatisfaction by 62% (OR=1.62; 95% CI: 1.08-2.44; $p=0.021$).

Although delayed emergency medical response demonstrated a positive association with dissatisfaction, the relationship did not reach statistical significance in the adjusted model (OR=1.41; 95% CI: 0.91-2.18; $p=0.118$).

Model diagnostics demonstrated acceptable calibration and predictive performance. The logistic regression model showed satisfactory goodness-of-fit according to the Hosmer-Lemeshow test ($p=0.47$), while Nagelkerke R^2 reached 0.28. Overall classification accuracy was 72.4%.

These findings indicate that dissatisfaction with cardiac care was primarily associated with infrastructural and organizational barriers rather than isolated emergency service factors. The multivariable associations between infrastructural determinants and patient dissatisfaction are presented in Figure 1.

The forest plot presents adjusted odds ratios with 95% confidence intervals derived from multivariable logistic regression analysis. Insufficient drug supply demonstrated the strongest independent association with dissatisfaction.

ROC analysis of the predictive model

Receiver operating characteristic (ROC) analysis demonstrated acceptable discrimination ability of the logistic regression model for predicting dissatisfaction with cardiac care.

The area under the curve (AUC) was 0.74 (95% CI: 0.69-0.79; $p<0.001$), indicating satisfactory predictive performance. At the optimal Youden index threshold, sensitivity reached 71%, while specificity was 68%.

The ROC analysis confirmed that the combined infrastructural indicators demonstrated moderate but clinically relevant predictive value for identifying dissatisfaction with specialized cardiac care.

Healthcare professional assessment of cardiac care quality

Healthcare professionals evaluated the overall quality of specialized cardiac care as moderate. The integral perceived quality index was 3.29 ± 1.04 on a five-point scale (Table 4).

Multivariate linear regression analysis demonstrated that adequacy of material and technical resources and digitalization level were independently

associated with higher perceived quality of cardiac care.

Material and technical resource adequacy demonstrated a significant positive association with perceived quality ($\beta=0.35$; $p=0.002$). Digitalization level showed the strongest independent relationship with quality assessment ($\beta=0.40$; $p=0.017$). Organizational coordination demonstrated a positive but statistically non-significant association ($\beta=0.18$; $p=0.094$).

The regression model explained 36% of variance in the perceived quality index ($R^2=0.36$; adjusted $R^2=0.33$; $F=11.24$; $p<0.001$). Multicollinearity diagnostics demonstrated acceptable values for all included predictors ($VIF<2$).

Table 4. Multivariate linear regression model for perceived quality of cardiac care (n=152).

Predictor	Standardized β	p-value
Material and technical resources	0.35	0.002
Digitalization level	0.40	0.017
Organizational coordination	0.18	0.094

These findings suggest that healthcare professionals primarily associated quality of specialized cardiac care with infrastructural capacity and digital integration rather than exclusively with clinical competencies.

Effect size analysis

Effect size analysis demonstrated that the identified disparities were predominantly concentrated in infrastructural components of healthcare accessibility (Table 5).

The association between rural residence and inadequate accessibility of high-technology medical care demonstrated a small-to-moderate effect size (Cramer's $V=0.21$). Drug supply disparity between urban and rural populations also demonstrated a clinically relevant effect size (Cramer's $V=0.18$). The

Table 5. Effect size estimates for major accessibility indicators.

Indicator	Effect size	Interpretation
Accessibility of HTMC	$V=0.21$	Small-to-moderate
Drug supply sufficiency	$V=0.18$	Small-to-moderate
Overall satisfaction	$d=0.32$	Small

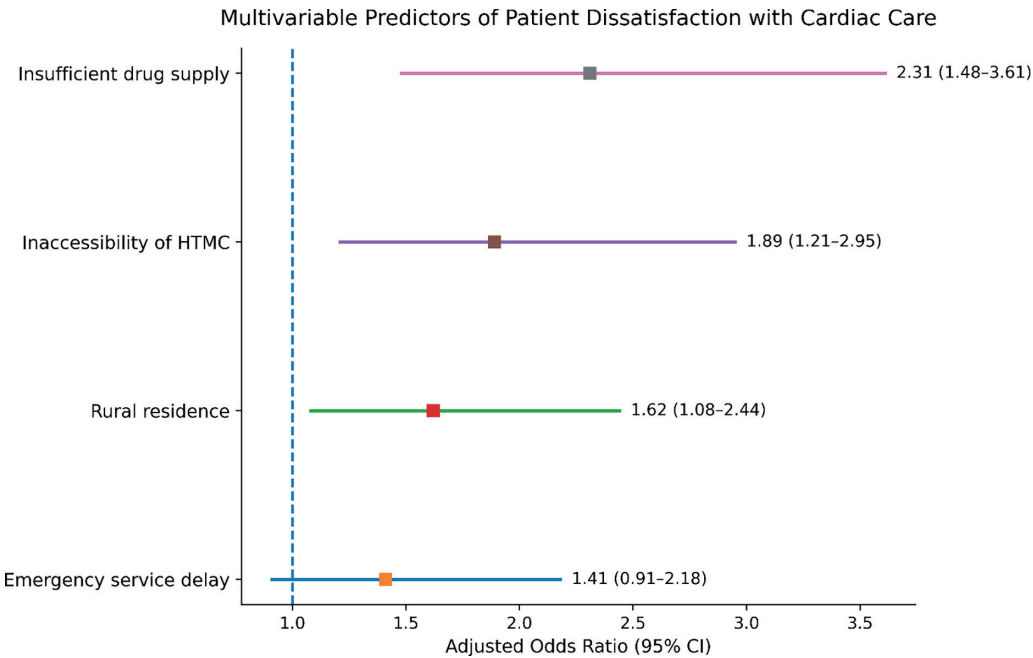


Figure 1. Adjusted odds ratios for infrastructural determinants associated with patient dissatisfaction with specialized cardiac care.

difference in overall satisfaction scores between urban and rural populations demonstrated a small but meaningful effect (Cohen's $d=0.32$).

Overall, the results indicate that urban-rural disparities in specialized cardiac care were driven primarily by infrastructural limitations, insufficient accessibility of high-technology services, and pharmaceutical supply instability rather than by differences in perceived clinical competence.

Discussion

Principal findings

The present study demonstrated substantial urban-rural disparities in accessibility and perceived quality of specialized cardiac care in Kazakhstan. The findings indicate that dissatisfaction with cardiac care was driven predominantly by infrastructural and organizational barriers rather than by isolated clinical factors. Insufficiency of pharmaceutical supply, limited accessibility of high-technology medical care, and rural residence emerged as the principal determinants associated with negative patient assessment of cardiac services. In parallel, healthcare professionals identified adequacy of material resources and digitalization level as the strongest determinants of perceived service quality.

Against the background of ongoing healthcare modernization in Kazakhstan, the observed disparities remain particularly important, as expansion of specialized cardiology services has not been accompanied by equivalent improvements in regional accessibility and continuity of care (19,20). Previous national studies similarly demonstrated persistent differences in cardiovascular morbidity, healthcare accessibility, and preventive strategies between urban and rural populations (21, 22). Our results extend these observations by showing that infrastructural inequality directly influences patient satisfaction and perceived quality of care.

Urban-Rural disparities in cardiac care accessibility

A marked disparity in access to high-technology cardiac care was observed between urban and rural

populations. Rural participants less frequently reported availability of advanced diagnostic procedures, specialized cardiac interventions, and timely access to medical services. Similar disparities have been consistently described in international cardiovascular literature (23-25).

Studies from Canada, Australia, and the United States demonstrated that rural populations experience delayed reperfusion therapy, lower accessibility of percutaneous coronary intervention, and higher cardiovascular mortality following acute myocardial infarction (23,24). Infrastructural barriers including transportation limitations, geographical remoteness, fragmentation of referral systems, and shortages of specialized personnel remain major contributors to these disparities (24,25).

Our findings are particularly relevant for Kazakhstan because of the country's large geographical territory and heterogeneous distribution of healthcare resources. Mukasheva et al. reported considerable regional variation in cardiovascular incidence and mortality across urban and rural populations of Kazakhstan and suggested that limited diagnostic infrastructure in rural areas may contribute to underdiagnosis and delayed treatment (6). Similar conclusions were reported by Glushkova et al., who demonstrated that cardiovascular prevention strategies in Kazakhstan remain predominantly tertiary-oriented, while secondary prevention and early risk detection remain insufficiently developed in rural populations (21).

An important observation of the present study is that rural residence remained independently associated with dissatisfaction even after adjustment for major infrastructural variables. This suggests that urban-rural inequality likely reflects not only organizational barriers, but also broader socioeconomic and contextual determinants including health literacy, transportation accessibility, delayed healthcare-seeking behavior, and workforce shortages.

Pharmaceutical accessibility as a determinant of patient satisfaction

One of the strongest findings of the present study was the association between insufficient pharmaceutical supply and dissatisfaction with cardiac care.

Patients reporting limited availability of prescribed medications demonstrated more than twofold higher odds of dissatisfaction with healthcare services.

This observation is clinically important because continuity of pharmacotherapy represents one of the major determinants of long-term cardiovascular outcomes. International evidence demonstrates that interruptions in cardiovascular drug supply are associated with increased rates of rehospitalization, recurrent cardiovascular events, and mortality (26,27). In low- and middle-income countries, pharmaceutical accessibility often reflects broader systemic problems including fragmented procurement mechanisms, regional inequities in healthcare financing, and unstable logistics chains (27).

The importance of medication adherence and continuity of therapy has also been emphasized in recent analyses of avoidable cardiovascular mortality in Kazakhstan. Junusbekova et al. demonstrated that despite reductions in cardiovascular mortality during the last decade, preventable and treatable mortality remain high, particularly in younger populations and rural regions (4). The authors additionally highlighted insufficient population health literacy and low adherence to treatment as important unresolved problems of the national healthcare system.

Taken together, these findings suggest that pharmaceutical accessibility may represent one of the most sensitive indicators of healthcare system functionality in geographically fragmented healthcare systems.

Role of digitalization and organizational infrastructure

Healthcare professionals participating in the present study consistently identified digitalization and adequacy of material and technical resources as the principal determinants of perceived quality of specialized cardiac care. These findings are highly consistent with contemporary international evidence regarding the growing role of digital health technologies in cardiovascular medicine.

Telecardiology, remote monitoring systems, electronic referral pathways, and hybrid rehabilitation models have demonstrated substantial potential for improving continuity of care and reducing geographical

barriers in underserved populations (28–30). Recent international studies additionally demonstrated that digital health integration may reduce avoidable hospitalizations and improve management of chronic cardiovascular diseases, particularly among patients residing far from tertiary healthcare centers (28).

The relevance of digitalization is especially important for Kazakhstan, where significant distances between rural settlements and specialized cardiology centers complicate timely access to care. Recent analyses of Kazakhstan's unified electronic healthcare system demonstrated active expansion of digital infrastructure and increasing availability of population-based registries for epidemiological monitoring (31). However, regional heterogeneity in implementation remains substantial.

Our findings indicate that digitalization should not be viewed exclusively as technological modernization, but rather as a strategic mechanism for reducing infrastructural inequality and improving healthcare equity in geographically heterogeneous healthcare systems.

Cardiac rehabilitation and continuity of care

Limited accessibility of cardiac rehabilitation services was also more frequently reported among rural participants. This finding is consistent with international evidence indicating that participation in cardiac rehabilitation contributes to better functional recovery, lower risk of recurrent cardiovascular events, and improved long-term survival (32). Nevertheless, accessibility of rehabilitation services remains substantially lower in rural populations because of transportation barriers, insufficient rehabilitation infrastructure, and shortage of specialized personnel (32,33). Recent data from Kazakhstan demonstrated that implementation of phase II cardiac rehabilitation under the compulsory insurance system improved cardiac function and clinical outcomes among patients with acute myocardial infarction; however, insufficient government funding continues to limit broader implementation of rehabilitation programs (34).

Given the growing burden of chronic cardiovascular diseases in Kazakhstan, expansion of outpatient, hybrid, and tele-rehabilitation programs may

therefore represent an important direction for reducing regional disparities in long-term cardiovascular outcomes. An important strength of the present study is the multidimensional analytical design integrating patient-reported accessibility indicators, healthcare professional assessments, infrastructural characteristics, and multivariable statistical modeling within a unified framework. Unlike many previous investigations focused primarily on epidemiological trends or clinical outcomes, the present study evaluated organizational determinants directly associated with perceived accessibility and quality of cardiac care.

Limitations

Several limitations should be taken into account when interpreting the findings of this study. The cross-sectional design does not allow causal relationships between infrastructural factors and patient dissatisfaction to be established. In addition, part of the data was based on self-reported assessments, which may be influenced by subjective perception and recall bias. Because participation was voluntary, the possibility of selection bias cannot be completely excluded.

The study was conducted within a single region of Kazakhstan, and therefore the findings may not fully reflect the situation in other regions with different demographic, socioeconomic, and healthcare characteristics. Furthermore, objective clinical outcomes, including mortality, rehospitalization, treatment adherence, and long-term cardiovascular complications, were not assessed within the current analysis.

Finally, although the regression models demonstrated acceptable predictive performance, inclusion of additional socioeconomic and geographic variables such as income level, transportation accessibility, and educational status could provide a more comprehensive understanding of factors influencing accessibility and perceived quality of cardiac care.

Despite these limitations, the study provides important evidence regarding infrastructural inequalities in cardiovascular healthcare in Kazakhstan and highlights several areas that may be relevant for future healthcare planning and system improvement.

Conclusion

The study demonstrated that important differences in accessibility and perceived quality of specialized cardiac care continue to exist between urban and rural populations in Kazakhstan. Rural patients more frequently reported difficulties related to access to high-technology cardiac services, availability of prescribed medications, and rehabilitation care. At the same time, healthcare professionals identified limitations in infrastructure and digital integration as major barriers affecting the quality and continuity of cardiovascular care.

The findings suggest that further improvement of cardiovascular outcomes in Kazakhstan will depend not only on the expansion of specialized cardiology services, but also on strengthening coordination between healthcare levels, improving accessibility of medical resources, and reducing regional disparities in care delivery. Particular attention should be directed toward rural populations, where organizational and infrastructural limitations remain most pronounced.

Ethic Approval: Ethical approval for the study was obtained from the Local Ethics Committee of Kazakhstan Medical University "KSPH" (Protocol No. IRB-193-2024/8 dated 06 May 2024). The study was conducted in accordance with the Declaration of Helsinki. All participants provided written informed consent prior to participation.

Conflict of Interest: Each author declares that there are no commercial or financial relationships that could be construed as a potential conflict of interest related to the submitted manuscript.

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